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Tactile braille dice

Abstract

A tactile die for playing a game, the die comprising a plurality of sides, wherein each side comprises a face having a braille character, each side comprises 3 or more raised border edges surrounding the face, and wherein one of the 3 or more raised border edges is an indicator edge configured to indicate an orientation of the face, the indicator edge comprising an indent along a length thereof.

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Background/Summary

(1) The invention is directed towards dice for game playing, the dice configured for use by those having limited or no sight.

BACKGROUND

(2) Certain games, for example role-playing games like Dungeons & Dragons, employ polyhedral

dice. Braille dice may be required for those having limited or no sight. Desired are dice for game playing that are easily read via touch, are comfortable to hold and pleasant to touch, are stable when resting on a side, and are of a proper size.

SUMMARY

(3) Accordingly, A tactile die for playing a game, the die comprising a plurality of sides, wherein each side comprises a face having a braille character, each side comprises 3 or more raised border edges surrounding the face, and wherein one of the 3 or more raised border edges is an indicator edge configured to indicate an orientation of the face, the indicator edge comprising an indent along a length thereof.

(4) Also disclosed is a set of tactile dice comprising at least two different die as described, the set comprising die having 4, 6, 8, 10, 10, 12, or 20 sides.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure described herein is illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, features illustrated in the figures are not necessarily drawn to scale. For example, the dimensions of some features may be exaggerated relative to other features for clarity. Further, where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.

(2) FIG. 1A-FIG. 1J show a front, top, left side perspective view, a front view, a rear view, a left side view, a right side view, a top view, a bottom view, a front, top, right side perspective view, a rear, bottom, left side perspective view, and a rear, bottom, right side perspective view of a 20-sided die, respectively, according to some embodiments.

(3) FIG. 2 shows a 12-sided die, according to an embodiment.

(4) FIG. 3A and FIG. 3B show different 10-sided die, according to some embodiments.

(5) FIG. 4 shows an 8-sided die, according to an embodiment.

(6) FIG. 5 shows a 6-sided die, according to an embodiment.

(7) FIG. 6 shows a 4-sided die, according to an embodiment.

DETAILED DISCLOSURE

(8) FIG. 1A shows a perspective view of 20-sided icosahedron die **100**, according to some embodiments. Die **100** comprises 20 faces **101**. When rolled, die **100** will come to rest on a single face, and a single opposite face, the face to be read, will face straight up. Each triangular face **101** comprises braille characters **102**, and raised border edges **103** and **104**. Raised border edge **104** is an indicator edge, comprising indent **105** along a length thereof. Border edge **104** comprises smooth downward-sloped edges **106**, which slope down to lowest point **107**, about flush with faces **101**. Characters **102** fit neatly in faces **101**, comprising no number-indicating “pound” sign (as traditionally found in braille numbering).

(9) FIG. 1B shows a front view of 20-sided icosahedron die **100**, according to an embodiment. A distance D1 between points of indent **105** where it begins to slope down to low point **107**, comprises about 52% of the entire length of an edge from corner **109** to corner **109**. A distance D2 of a length of low point **107** is about 18% of the entire length from corner **109** to corner **109**, or about 33% of D1. Raised border edges **103** and **104** meet at corners **109**. Corners **109** have a substantially identical size and shape. Corners **109** have increased surface areas **108**, which are small enough not to crowd braille characters **102** and which provide for stability when die **100** is at rest on a face. An increased surface area **108** may be indicated for example, where a triangular face comprises rounded corners as in FIG. 1B. FIG. 1C through FIG. 1J provide further views of die **100**.

(10) FIG. 2 shows 12-sided dodecahedron die **200**, according to an embodiment. Die **200**

comprises 12 faces **201**, braille characters **202**, and raised border edges **203** and **204**. Indicator border edges **204** comprise indents **205**. Corners **209** comprise increased surface areas **208** which take up some surface area from pentagonal faces **201**.

(11) FIG. 3A and FIG. 3B show different 10-sided pentagonal trapezohedron die **300a** and **300b**, according to some embodiments. Die **300a** and **300b** comprise 10 faces **301**, braille characters **302**, and raised border edges **303** and **304**. Indicator edges **304** are not linear, but bent or curved. Raised border edges **303**, like other border edges shown in the figures, are linear. Each side shares part of an indicator edge **304** with two other sides. Die **300a** comprises blunted corners **310**, while die **300b** comprises sharp corners **311**, thus differentiating the two die. Scoop **305** is centered along edges **304**.

(12) FIG. 4 shows 8-sided octahedron die **400**, according to an embodiment. Die **400** comprises 8 faces **401**, braille characters **402**, and raised border edges **403** and **404**. Indicator border edges **404** comprise indents or scoops **405**. Each triangular side comprises 2 raised edges **403** and one indicator edge **404**. Every side shares an indicator edge **404**. Corners **409** comprise increased surface areas **408**, taking up some surface area of triangular faces **401**.

(13) FIG. 5 shows 6-sided cube die **500**, according to an embodiment. Die **500** comprises 6 faces **501**, braille characters **502**, and raised border edges **503** and **504**. Indicator edges **504** comprise indents or scoops **505**. Each square side comprises 3 edges **503** and one indicator edge **504**. Each side shares an indicator edge **504**. Corners **509** comprise increased surface areas **508**, taking up some surface area of square faces **501**.

(14) FIG. 6 shows 4-sided die **600**, according to an embodiment. Die **600** comprises 4 faces **601**, braille characters **602**, and raised border edges **603** and **604**. Indicator edges **604** comprise indents or scoops **605**. Each rectangular side comprises 3 edges **603** and one indicator edge **604**. Each side shares an indicator edge **604**. Corners **609** comprise increased surface areas **608**, taking up some surface area of rectangular faces **601**. Four-sided die **600** contains two rounded, dome-shaped sides **625** having no characters. Sides **625** are opposite each other.

(15) Following are indicator numbers of features in the figures. **100 200 300a 300b 400 500 600** die **101 201 301 401 501 601** die face **102 202 302 402 502 602** braille characters **103 203 303 403 503 603** raised border edge **104 204 304 404 504 604** raised border edge—indicator edge **105 205 305 405 505 605** edge indent **106 206 306 406 506 606** downward slope of edge **107 207 307 407 507 607** indicator edge low point **108 208 308 408 508 608** corner increased surface area **109 209 409 509 609** corner **625** rounded side **310** blunted corner **311** sharp corner

(16) In some embodiments, an indent positioned along a length of a raised indicator edge may be substantially centered along a length of the edge. In other embodiments, an indent may be positioned off-center along a length of the edge. An indent may be a notch having any shape, for instance a triangular, rectangular, square, arc shape, etc. In some embodiments, an indent may be referred to as a “scoop” or a “divot”. An indent may have sharp edges and/or corners. In other embodiments, an indent may have smooth edges. In some embodiments, an indent may have smooth, downward-sloped edges, which slope down to a lowest point. In some embodiments, a lowest point may be flush or about flush with a face surface. In other embodiments, a lowest point may be positioned at a point above a face surface.

(17) A face is generally planar and contains braille characters. A face is bordered by raised border edges, wherein one or more of the raised edges are indicator edges. In general, a side will comprise a single indicator edge. In some cases, a side (a minority of sides) will contain only indicator edges (thus not being indicator edges). In some embodiments, all edges are generally linear. In other embodiments, some edges may be non-linear, for example bent or curved. In some embodiments, an indicator edge may be bent or curved.

(18) In some embodiments, an indicator edge may comprise a single indent. In other embodiments, an indicator edge may comprise 2, 3, 4, or more indents. In some embodiments, two sides of a die may share a single indicator edge. An indicator edge allows one to determine an orientation of a

die, for example may indicate which way is up, for example how to read a 6 vs. 9.

(19) In an embodiment, the present braille characters are configured to be interpreted as numbers. Typically, braille characters to be interpreted as numbers use letters A-J to represent numbers 1-10, and are preceded by a “pound” symbol to indicate it should be read as a number. For example “#AH” would indicate the number 18, #C would indicate 3, etc. In some embodiments, present braille characters employ no “pound symbol”, and may simply use a corresponding alphabet letter to indicate a number. For example, S may represent 19, P may represent 16, C may represent 3, etc. A present number indicating system may allow for more space on a die side, thus allowing for corners having increased surface area.

(20) In some embodiments, an indent, notch, or scoop may comprise a length of from any of about 3.0 mm, about 3.5 mm, about 4.0 mm, about 4.5 mm, about 5.0 mm, about 5.5 mm, about 6.0 mm, or about 6.5 mm, to any of about 7.0 mm, about 7.5 mm, about 8.0 mm, about 8.5 mm, about 9.0 mm, about 9.5 mm, about 10.0 mm, about 10.5 mm, about 11.0 mm, or more, measured from inflection points where the indent begins to angle downwards, along a length of an edge.

(21) In some embodiments, an indent, notch, or scoop length, measured between inflection points, may make up from any of about 15%, about 18%, about 20%, about 23%, about 26%, about 29%, about 32%, or about 35%, to any of about 38%, about 41%, about 44%, about 47%, or more, of an entire length of an edge from corner to corner.

(22) In some embodiments, a length of an indent, notch, or scoop low point may comprise a length of from any of about 0.5 mm, about 1.0 mm, about 1.3 mm, about 1.6 mm, about 1.9 mm, or about 2.2 mm, to any of about 2.5 mm, about 2.8 mm, about 3.1 mm, about 3.4 mm, about 3.7 mm, about 4.0 mm, about 4.3 mm, about 4.7 mm, about 5.0 mm, about 5.3 mm, about 5.6 mm, about 5.9 mm, about 6.2 mm, about 6.5 mm, about 6.8 mm, or more, measured along a low point edge surface.

(23) In some embodiments, a length of an indent, notch, or scoop low point, measured along a low point edge surface, may make up from any of about 10%, about 13%, about 16%, about 19%, about 22%, or about 25%, to any of about 28%, about 31%, about 34%, about 37%, or more, of an entire length of an edge from corner to corner.

(24) In some embodiments, a height of a braille character from a face may be about the same as a height of a border edge. In other embodiments, a braille character height may be slightly less than a height of a border edge. In some embodiments, a braille character height may be from about 40%, about 45%, or about 50%, to any of about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, or about 100% of the height of a border edge.

(25) In some embodiments, a die side comprises a die face and border edges surrounding the face. Upon rolling a die, it may come to rest on raised edges. In some embodiments, an increased corner surface area provides for increased stability when a die is at rest on its edges.

(26) It has been found that present die are comfortable to the touch, are easy to read, and are stable when at rest on a side thereof.

(27) The term “adjacent” may mean “near” or “close-by” or “next to”.

(28) The term “towards” in reference to a point of attachment, may mean at exactly that location or point or, alternatively, may mean closer to that point than to another distinct point, for example “towards a center” means closer to a center than to an edge.

(29) The term “like” means similar and not necessarily exactly like. For instance “ring-like” means generally shaped like a ring, but not necessarily perfectly circular.

(30) The articles “a” and “an” herein refer to one or to more than one (e.g. at least one) of the grammatical object. Any ranges cited herein are inclusive. The term “about” used throughout is used to describe and account for small fluctuations. For instance, “about” may mean the numeric value may be modified by $\pm 0.05\%$, $\pm 0.1\%$, $\pm 0.2\%$, $\pm 0.3\%$, $\pm 0.4\%$, $\pm 0.5\%$, $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$, $\pm 6\%$, $\pm 7\%$, $\pm 8\%$, $\pm 9\%$, $\pm 10\%$ or more. All numeric values are modified by the term “about” whether or not explicitly indicated. Numeric values modified by the term “about” include the specific identified value. For example “about 5.0” includes 5.0.

- (31) The term “substantially” is similar to “about” in that the defined term may vary from for example by $\pm 0.05\%$, $\pm 0.1\%$, $\pm 0.2\%$, $\pm 0.3\%$, $\pm 0.4\%$, $\pm 0.5\%$, $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$, $\pm 6\%$, $\pm 7\%$, $\pm 8\%$, $\pm 9\%$, $\pm 10\%$ or more of the definition; for example the term “substantially perpendicular” may mean the 90° perpendicular angle may mean “about 90° ”. The term “generally” may be equivalent to “substantially”.
- (32) Features described in connection with one embodiment of the disclosure may be used in conjunction with other embodiments, even if not explicitly stated.
- (33) Embodiments of the disclosure include any and all parts and/or portions of the embodiments, claims, description and figures. Embodiments of the disclosure also include any and all combinations and/or sub-combinations of embodiments.

Claims

1. A tactile die for playing a game, the die comprising a plurality of sides, wherein each side comprises a face to be read having a braille character, each side comprises 3 or more raised border edges surrounding the face, upon being rolled, the die is configured to rest on the raised border edges, the die will rest on a single side, and a single opposite side comprising the face to be read will face straight up, and a single one of the 3 or more raised border edges surrounding each face is an indicator edge configured to indicate an orientation of the face to be read, the indicator edge comprising an indent along a length thereof.
2. The tactile die according to claim 1, wherein the braille character is configured to be read as a number.
3. The tactile die according to claim 2, wherein the braille character is not preceded by a number-indicating pound sign.
4. The tactile die according to claim 1, wherein the indent is substantially centered along a length of the raised border edge.
5. The tactile die according to claim 1, wherein the indent comprises smooth downward-sloped edges.
6. The tactile die according to claim 1, wherein a lowest point of the indent is flush or about flush with the face.
7. The tactile die according to claim 1, wherein any two sides share the single indicator edge.
8. The tactile die according to claim 1, comprising 4, 6, 8, 10, 12, or 20 sides.
9. The tactile die according to claim 1, wherein the indicator edges are non-linear.
10. The tactile die according to claim 1, wherein each side comprises 3 raised border edges and a single indicator edge.
11. The tactile die according to claim 1, wherein each side comprises 4 raised border edges and a single indicator edge.
12. The tactile die according to claim 1, wherein each side comprises 5 raised border edges and a single indicator edge.
13. The tactile die according to claim 1, wherein the raised border edges meet at corners, and wherein the corners have an increased surface area.
14. The tactile die according to claim 1, wherein the raised border edges meet at corners, and wherein the corners are substantially identical in shape.
15. The tactile die according to claim 1, wherein the indicator edges are linear.
16. A set of tactile dice comprising at least two different die according to claim 1, the set comprising die having 4, 6, 8, 10, 12, or 20 sides.
17. The set of tactile dice according to claim 16, comprising 7 different die having 4, 6, 8, 10, 10, 12, and 20 sides.
18. The set of tactile die according to claim 17, wherein the two different die having 10 sides are differentiated by their corners.

19. The set of tactile die according to claim 18, wherein one of the two die having 10 faces comprises blunted corners.
